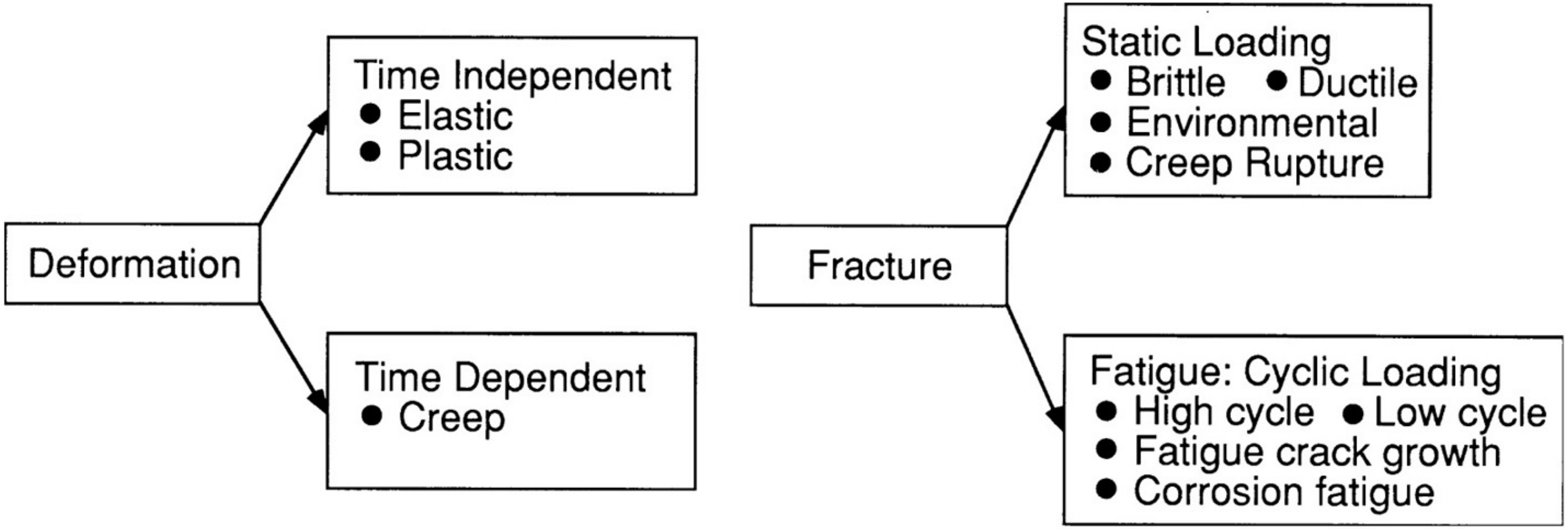
Upcoming Due Dates

- Module 3 – Prelab due today
- Module 2 – Worksheet Due Tuesday
  - Submit to Canvas before class
  - Bring copy to class for notes
- Module 2 Video – Due Sunday, 2/15

CH 6 & 7 Properties of Solid Materials
Mechanical Behavior of Materials



Mechanical Behavior of Materials: Engineering Methods for Deformation, Fracture, and Fatigue, Third Edition,
by Norman E. Dowling. ISBN 0-13-186312-6.

Today's Lecture

We are here!

3. Processing → 1. Structure → 2. Material Properties ↓

4. Performance

<u>Addressed in other courses</u>:

ENGR 2231: Strength of Materials

ME 3121: Design of Machine Components

CEE 3411: Structural Analysis

Why don't these break?

Automobile Drive Shaft

Golden Gate Bridge

Running Limbs

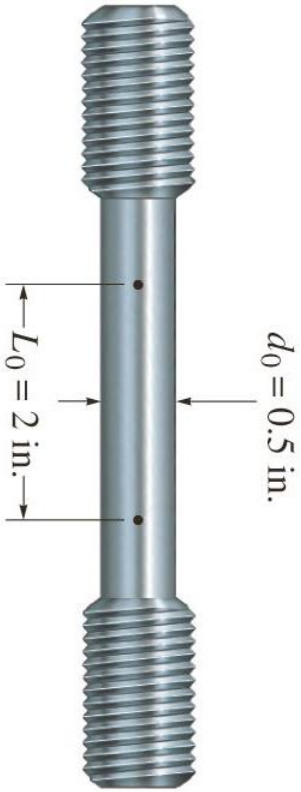
Skyscrapers (Burj Khalifa)

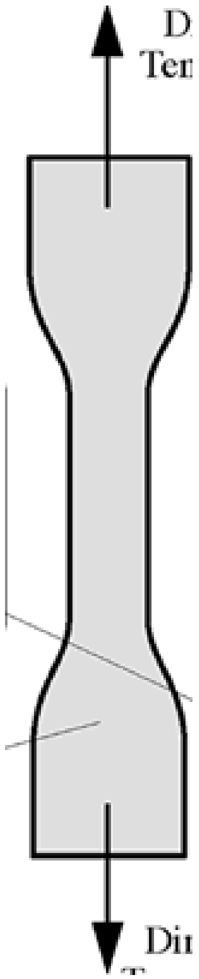
Engineers use material properties and the principles of math and science to predict the future!

What properties should be defined for a material for this application?

Many Material Properties are determined by a Tension Test

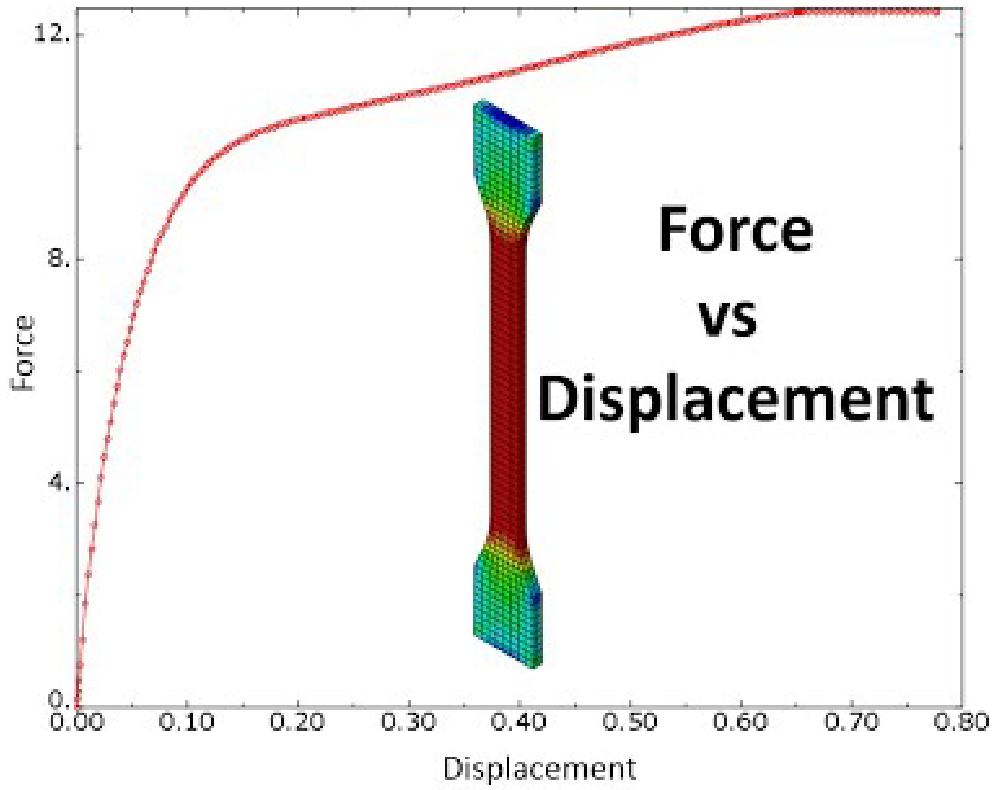
“uniaxial stress-strain response”





Mechanics of materials, R. C. Hibbeler

Force vs. Displacement Graph



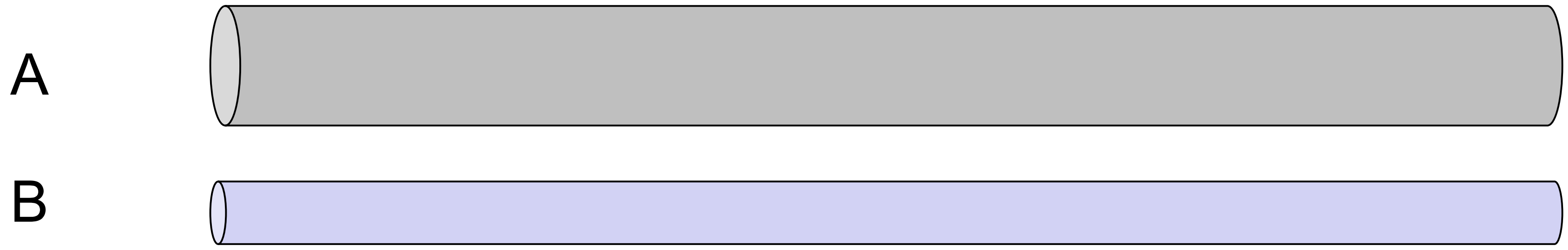
Hooke's Law:

$$F = kx$$

- A tensile testing machine measures Force and displacement
- Do force and displacement make good material properties?

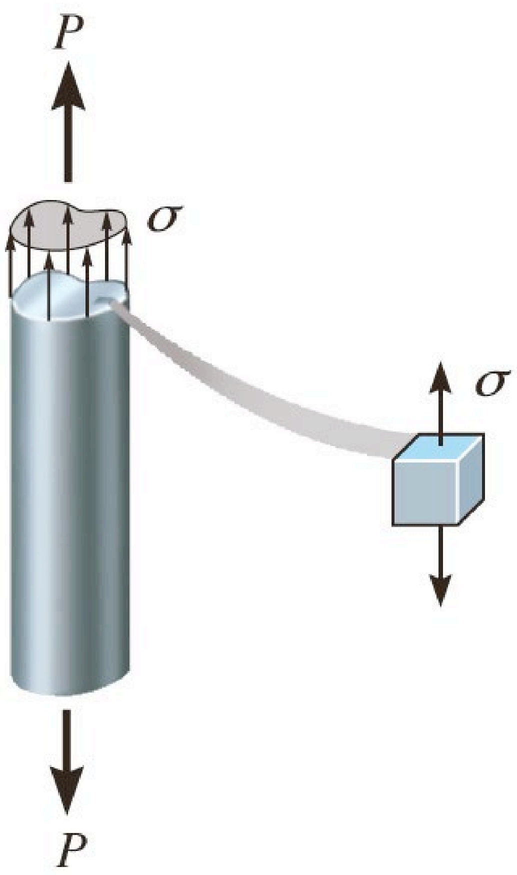
Does force make a good material property?

We have two AISI 1020 Steel Rods. Which will support more force before breaking?



- The amount of force before breaking is dependent on both the material and geometry
- We define properties that are <u>independent of geometry</u>, and thus inherent to the material itself

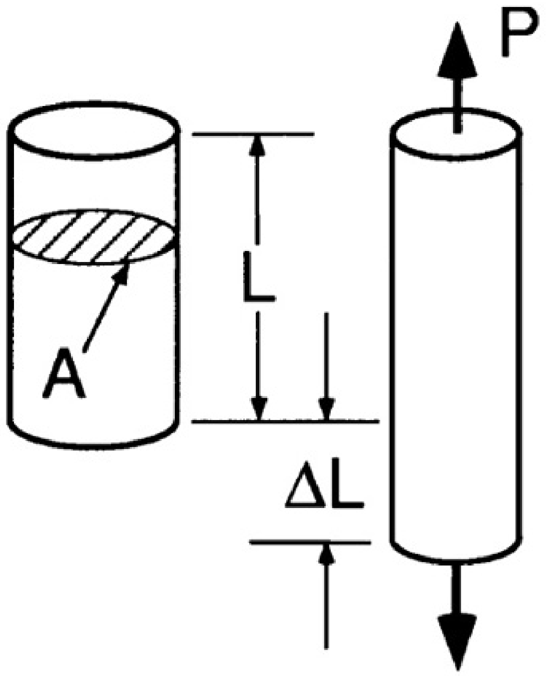
Engineering Stress (Axial Loading)



$$\sigma = \frac{P}{A_o} = \frac{\text{Applied Force}}{\text{Original Area of the cross section}}$$

- σ is used to represent “normal stresses”
- For axial loading, the normal stress is same across the entire cross section.
- Stress has units of :
  - Pounds per square inch: psi or kpsi
  - Pascals: Pa, kPa, MPa

Engineering Strain (Axial Loading)

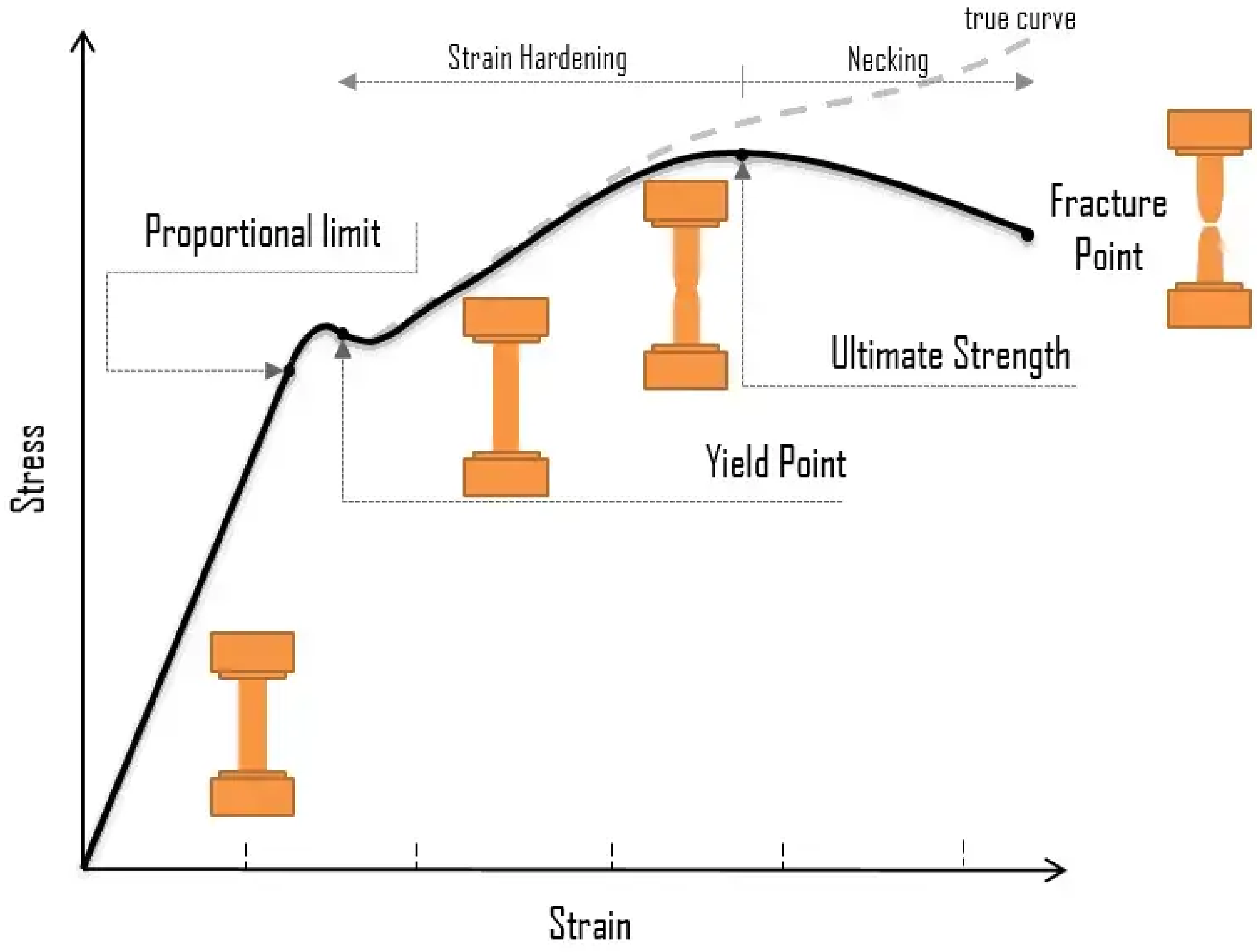


$$\varepsilon = \frac{\Delta l}{l_o} = \frac{\text{Change in Length}}{\text{Original Length}}$$

$$\Delta l = l - l_o \;\rightarrow \text{Current Length} \; - \text{Orginal Length}$$

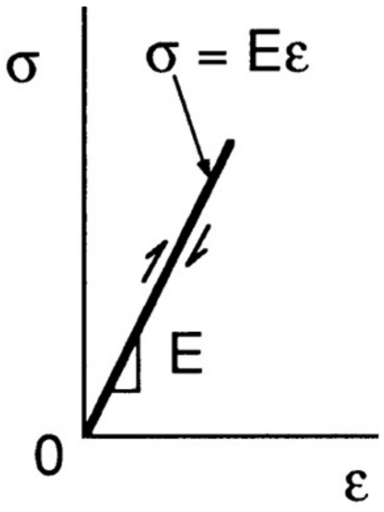
- ε is used to represent "normal strain"
- Strain is unit less and often very small
- 10^{-6} can be reported in units of με (micro strain)
  - Ex: $5.82 \times 10^{-6} = 5.82\ \mu\varepsilon$

The Stress-Strain Curve



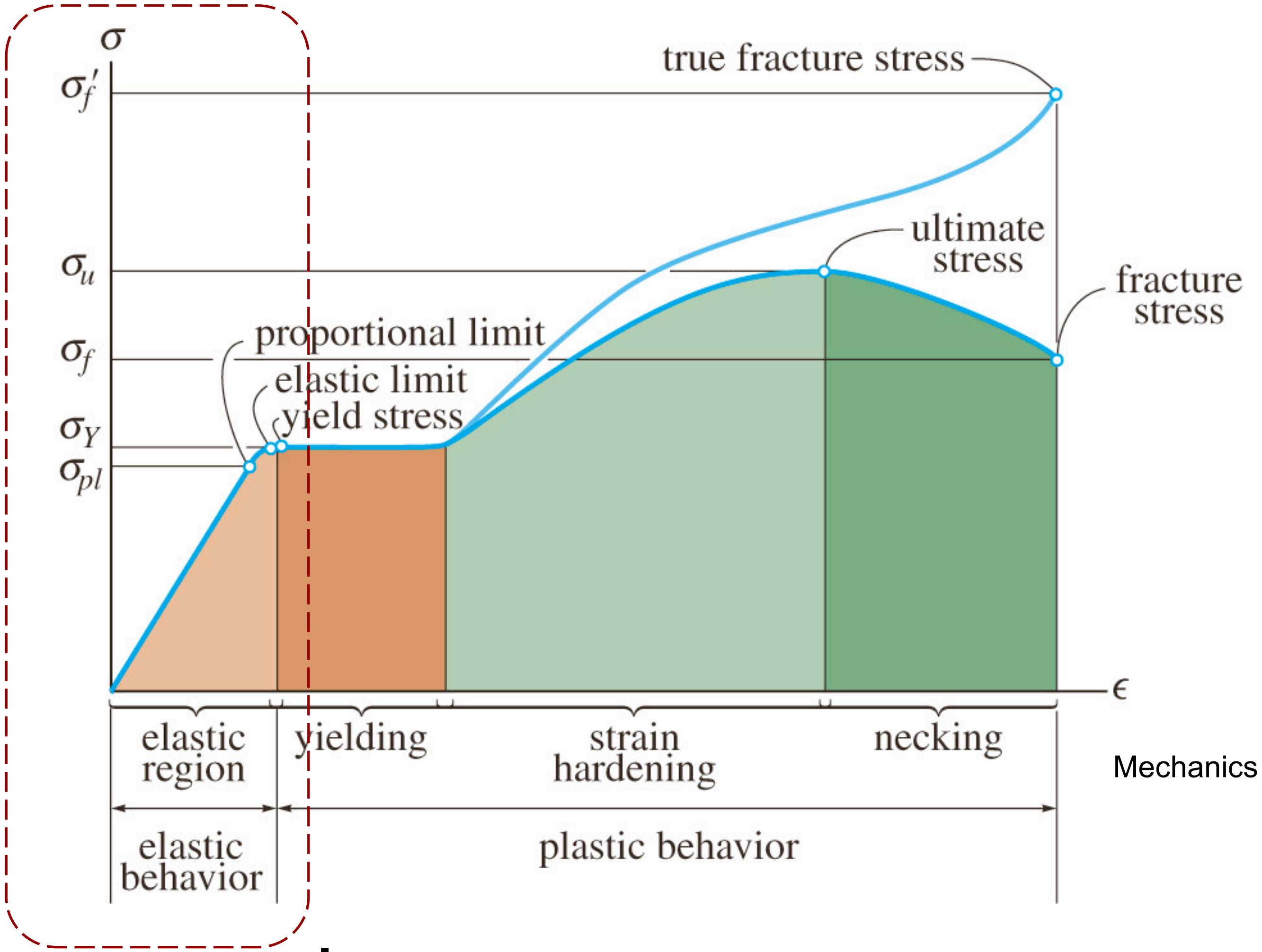
Inputting the geometry (cs shape, area, length, etc) allows for the calculation of the stress-strain curve

The Elastic Region:



Hooke's Law:

$$\sigma = E\varepsilon$$

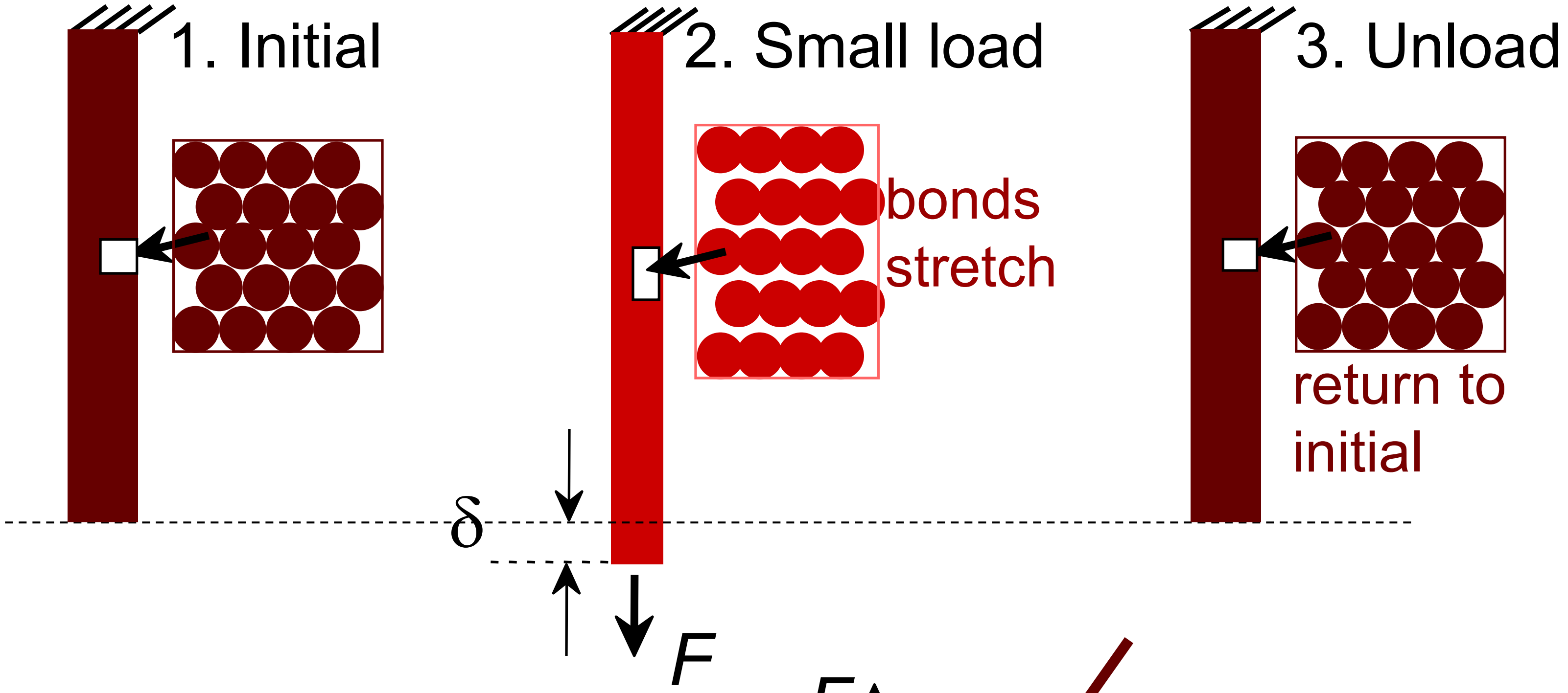


Mechanics of materials, R. C. Hibbeler

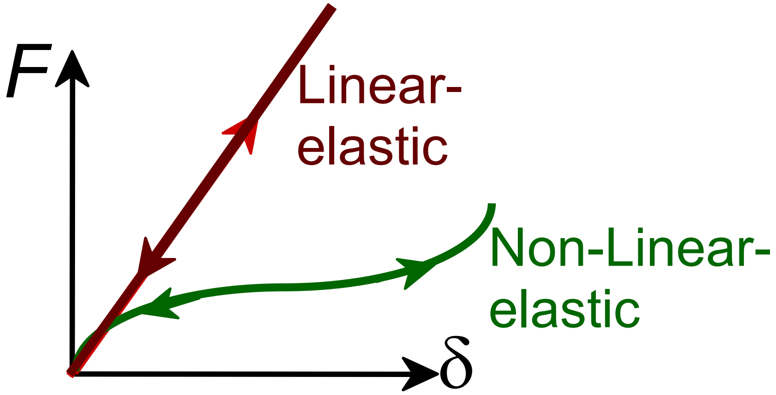
E: modulus of elasticity [also: Young's modulus, tensile elasticity]

a material property that describes its stiffness and is therefore one of the most important properties in engineering design.

Elastic Deformation



Elastic means reversible!



Elastic Modulus, E, is determined by bonding forces

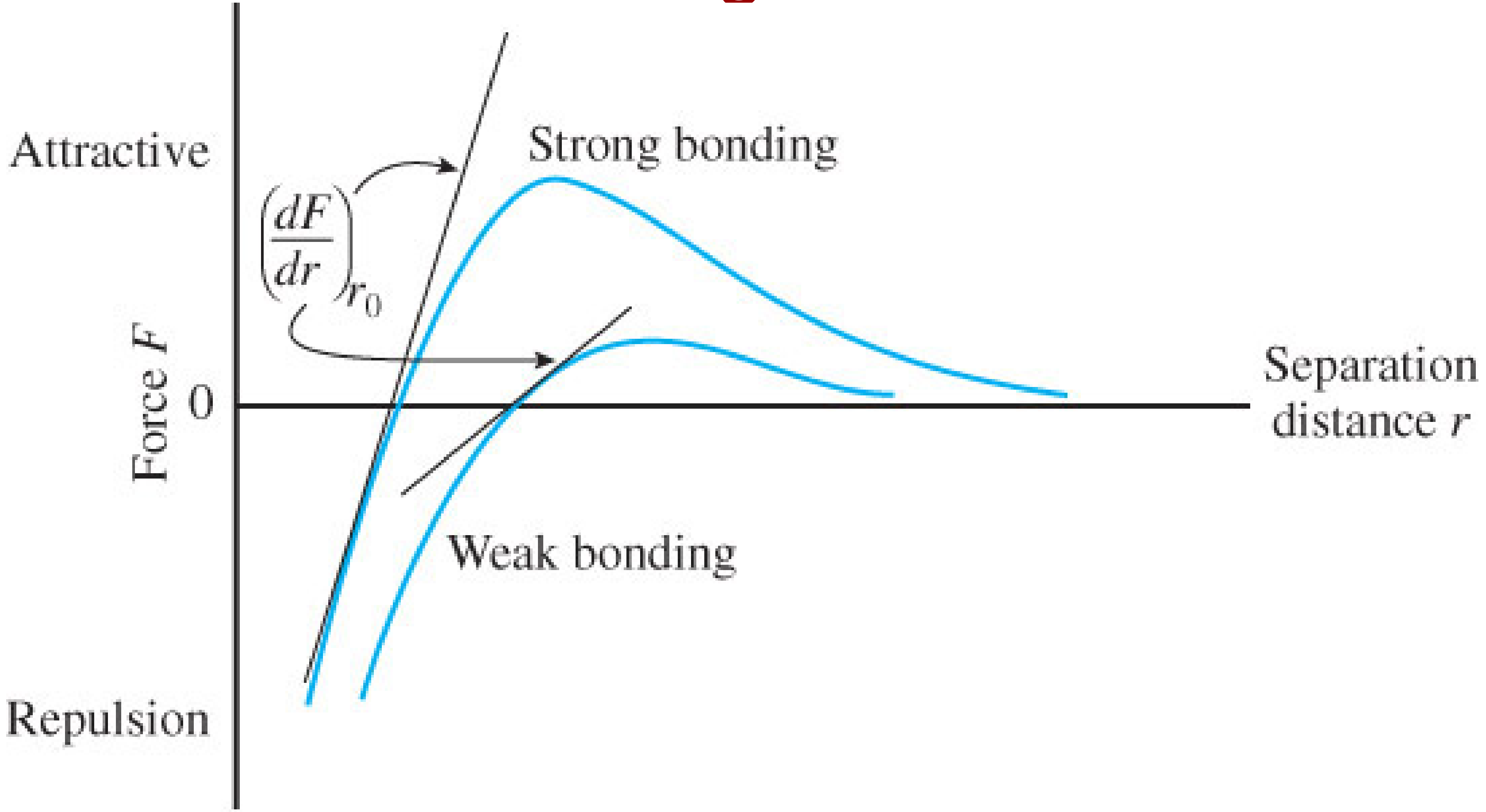
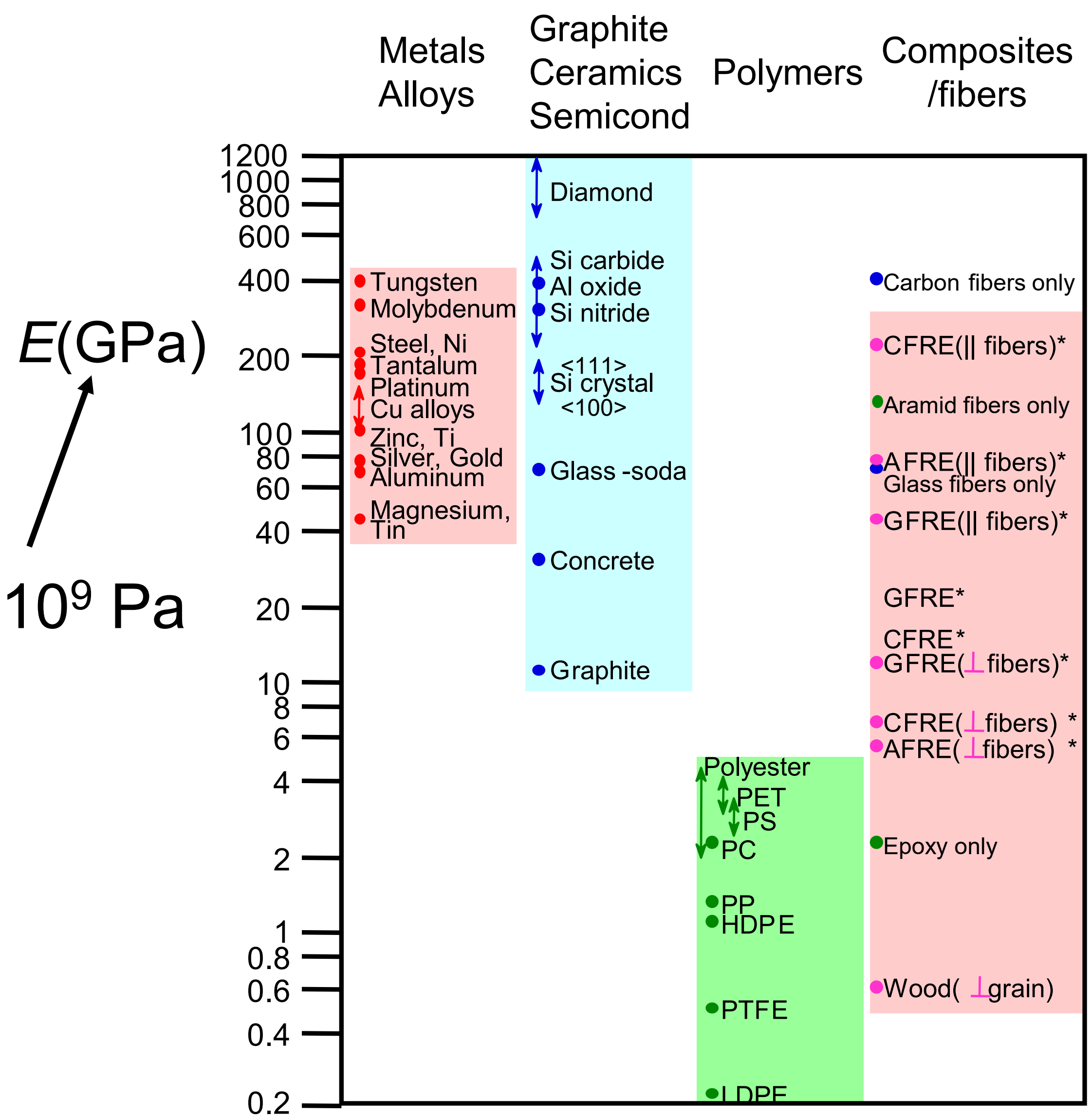


Figure 7.4

Young's Moduli: Comparison



Based on data in Table B.2, *Callister & Rethwisch 3e.* Composite data based on reinforced epoxy with 60 vol% of aligned carbon (CFRE), aramid (AFRE), or glass (GFRE) fibers.

Elastic Region: Using Hooke's Law

You can find the axial force (P) applied if you know the material (E) and new length (*l*) and original length (l_o):

$$\sigma = E\varepsilon$$

$$\frac{P}{A} = E\,\frac{\Delta l}{l_o}$$

$$P = EA\frac{l - l_o}{l_o}$$

You can find the original length (l_o) applied if you know the material (E) and final length (*l*) and force (P):

$$\sigma = E\varepsilon$$

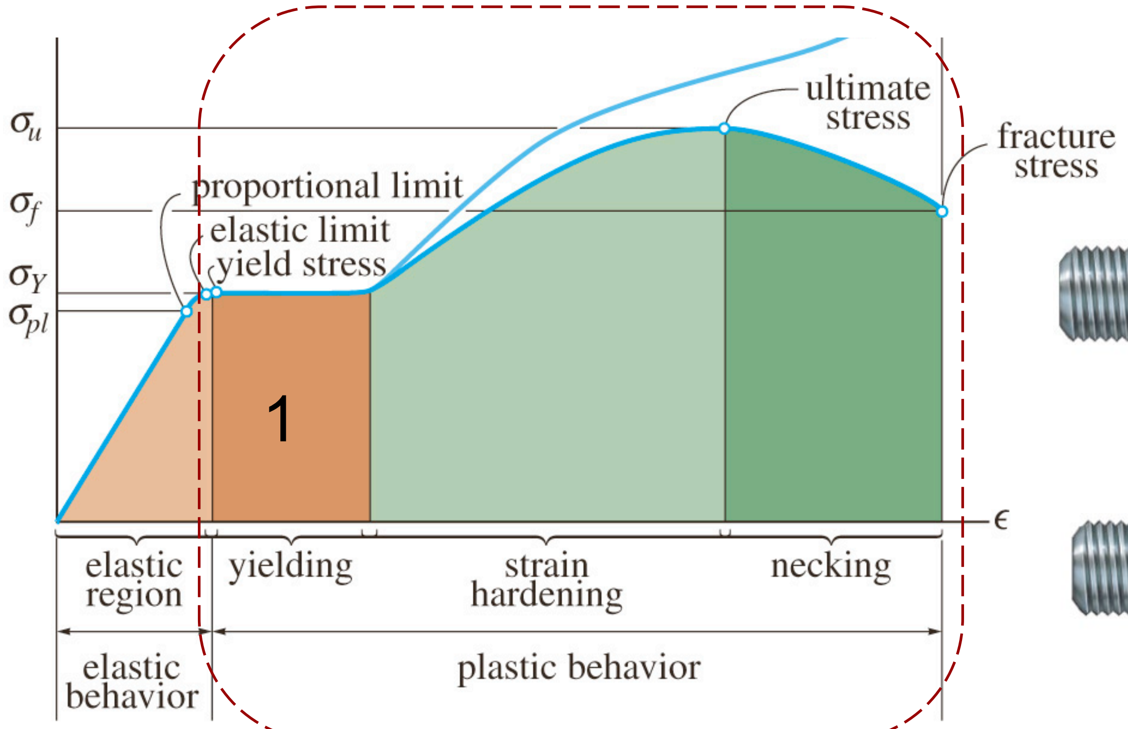
$$\frac{P}{A} = E\,\frac{\Delta l}{l_o}$$

Sometimes this format:

$$\Delta l = \frac{P l_o}{AE} \quad \rightarrow \quad \delta = \frac{PL}{AE}$$

$$l = \frac{P l_o}{AE} + l_o$$

Plastic Region



Conventional and true stress-strain diagrams

Necking

Failure of a ductile material

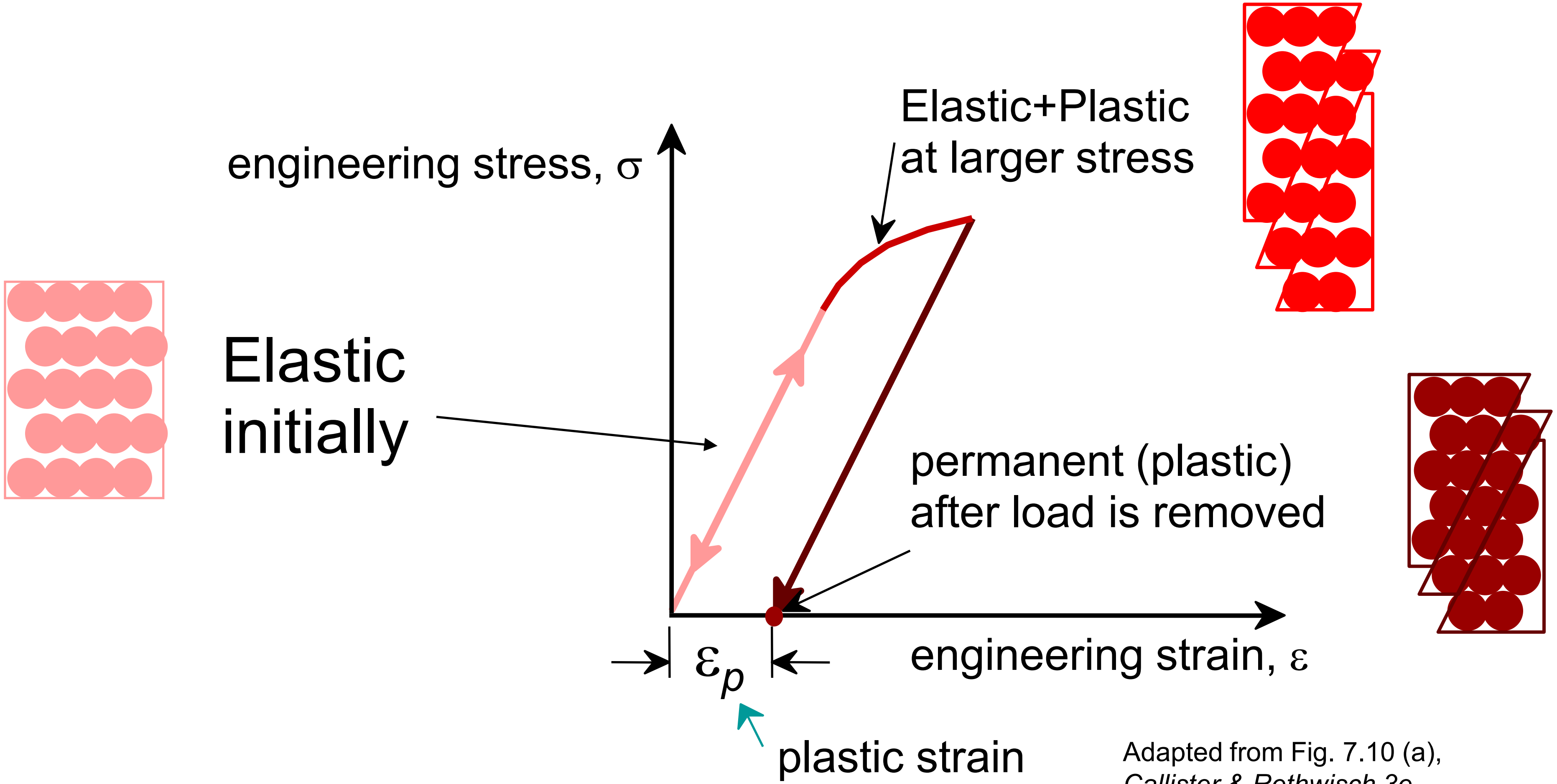
1. Yielding: A slight increase in stress above the elastic limit will result in a breakdown of the materials and cause it to deform permanently

Yield stress (or Yield strength): the stress that causes yielding. This is a measure of strength for ductile materials

Plastic deformation = permanent deformation.

Plastic (Permanent) Deformation

(at lower temperatures, i.e. $T < T_{melt}/3$)



Adapted from Fig. 7.10 (a),
Callister & Rethwisch 3e.

Plastic Deformation caused by Dislocation Motion

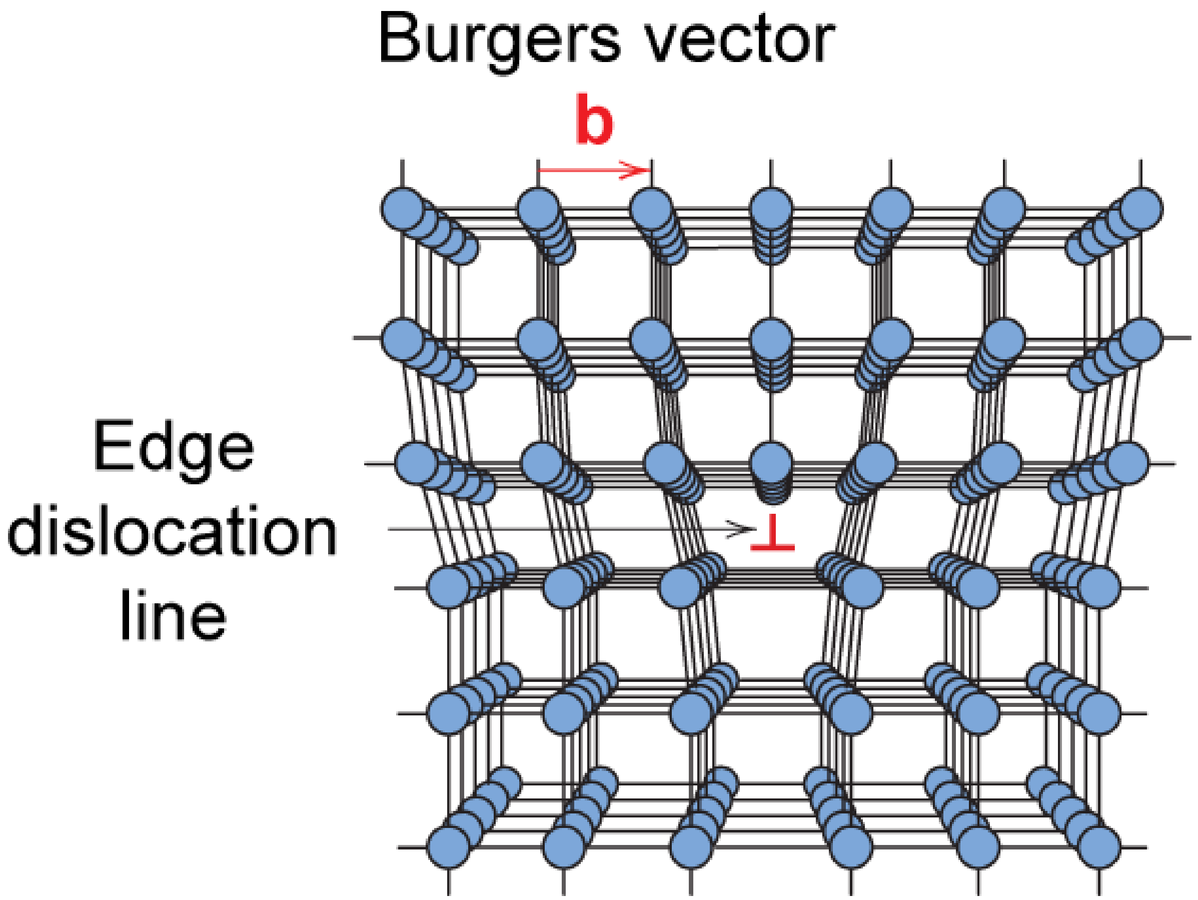
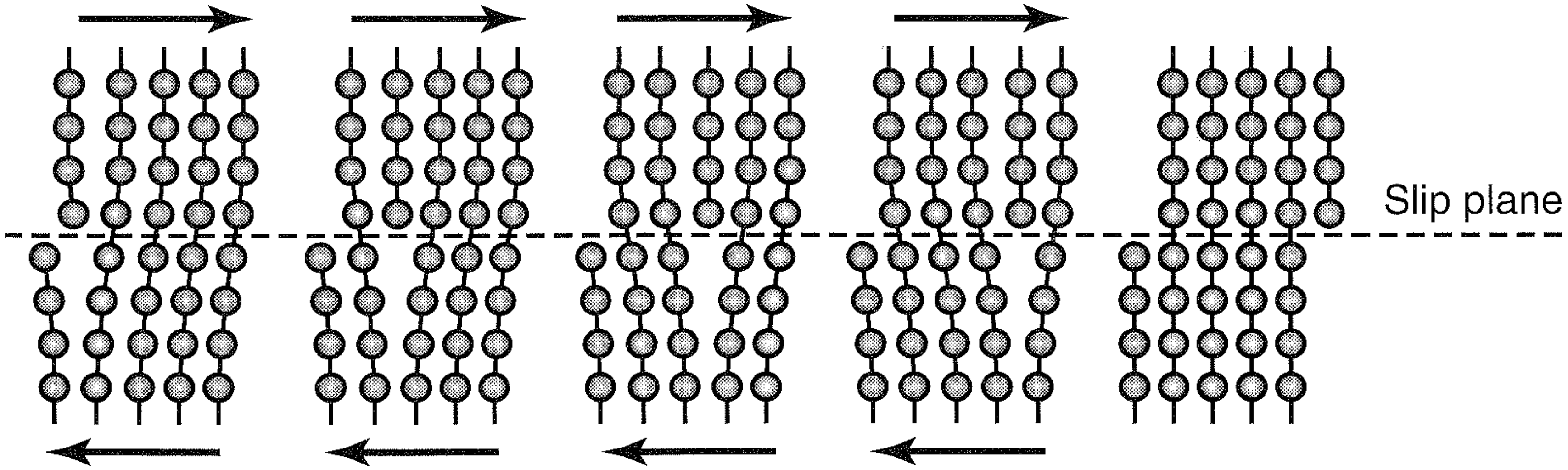


Fig. 5.8, *Callister & Rethwisch 3e.*

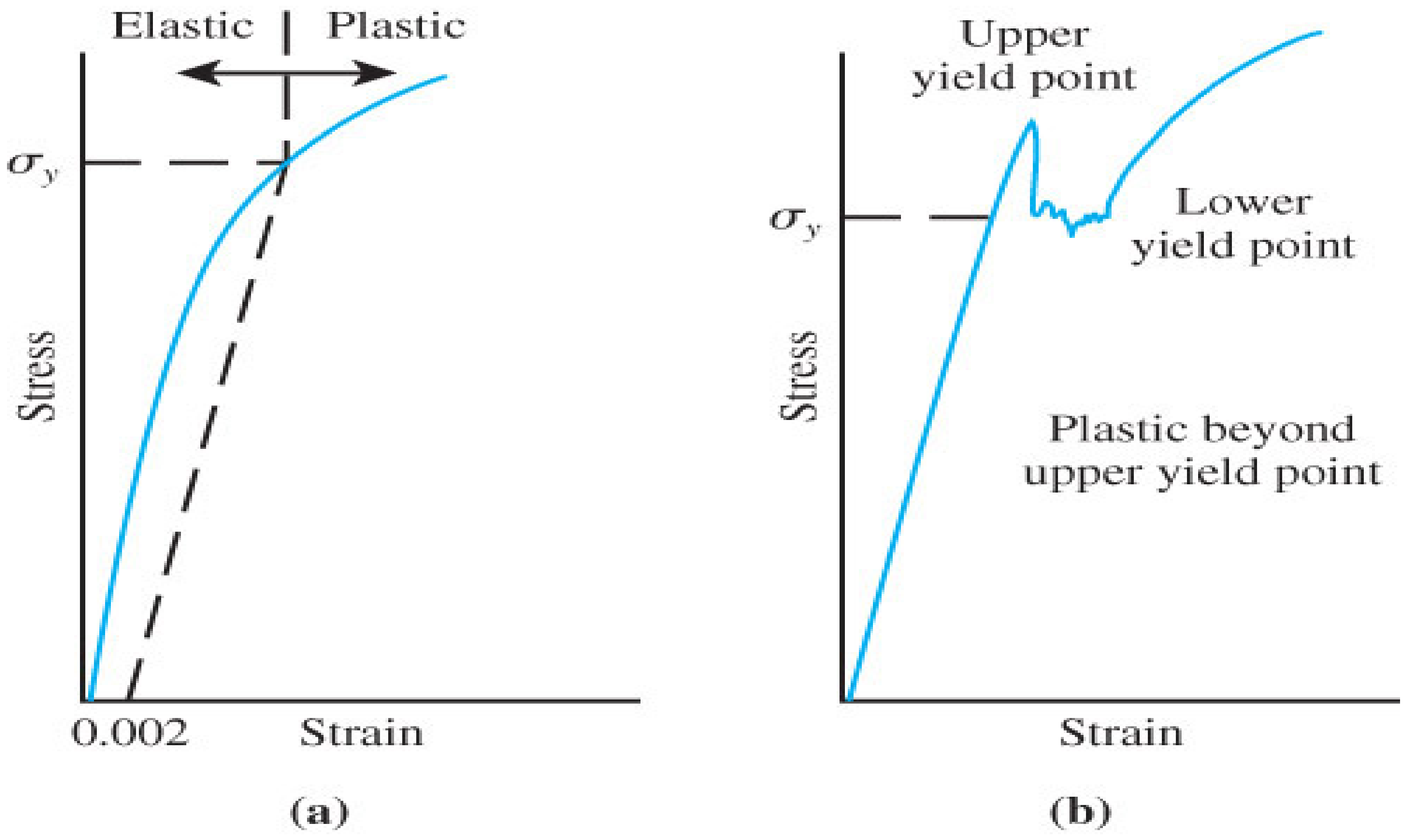
- Dislocation motion requires the successive bumping of a half plane of atoms (from left to right here).
- Bonds across the slipping planes are broken and remade in succession.

Plastic deformation occurs along a slip plane.

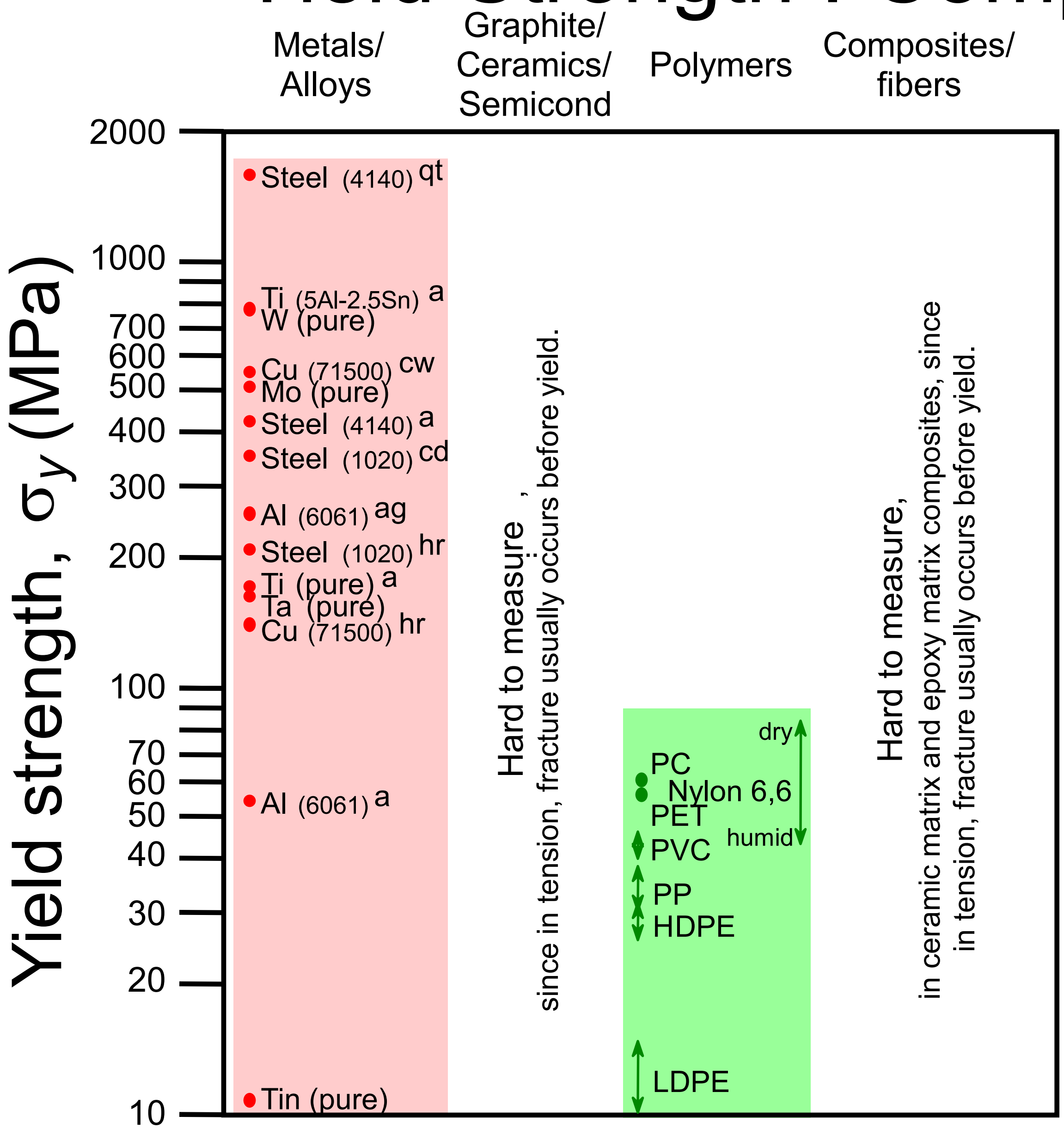


Kalpakjian and Schmid, *Manufacturing Engineering and Technology*, 5e.

0.2% Offset Yield Stress (a)
Upper and Lower Yield Point (b)



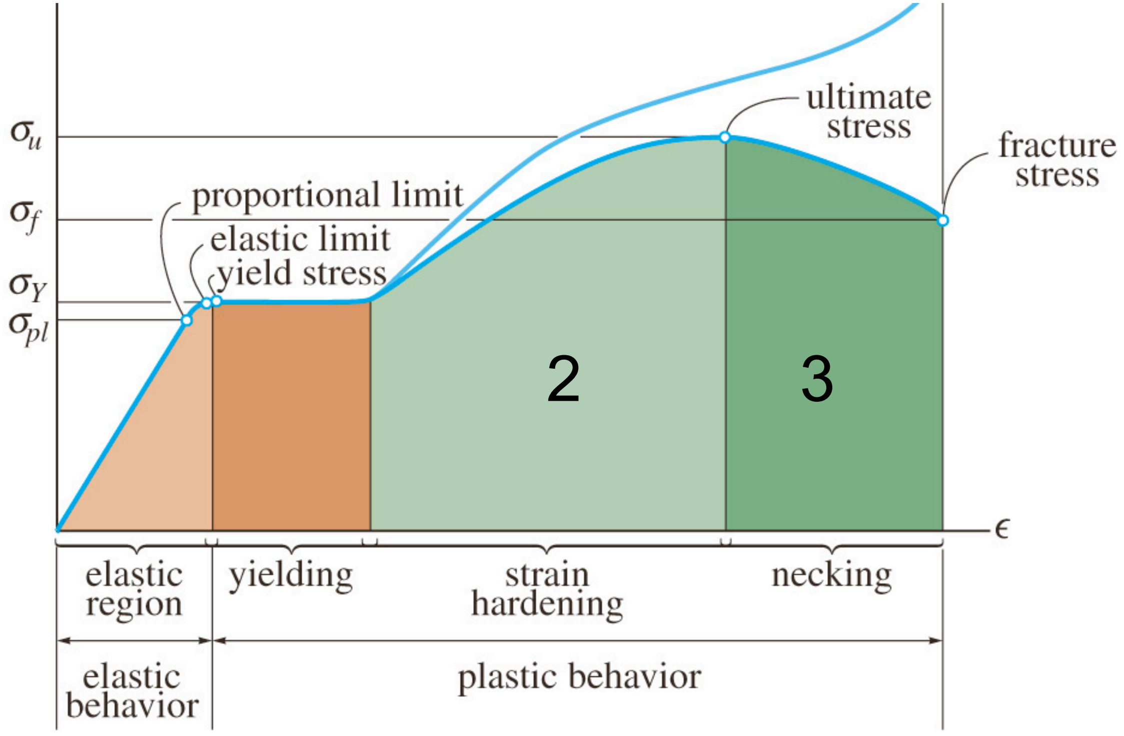
Yield Strength : Comparison



Room temperature values

Based on data in Table B.4, *Callister & Rethwisch 3e.*

a = annealed
hr = hot rolled
ag = aged
cd = cold drawn
cw = cold worked
qt = quenched & tempered



Conventional and true stress-strain diagrams

$$percent \quad elongation = \frac{L_f - L_o}{L_o}$$

$$percent \quad reduction \quad of \quad area = \frac{A_f - A_o}{A_o}$$

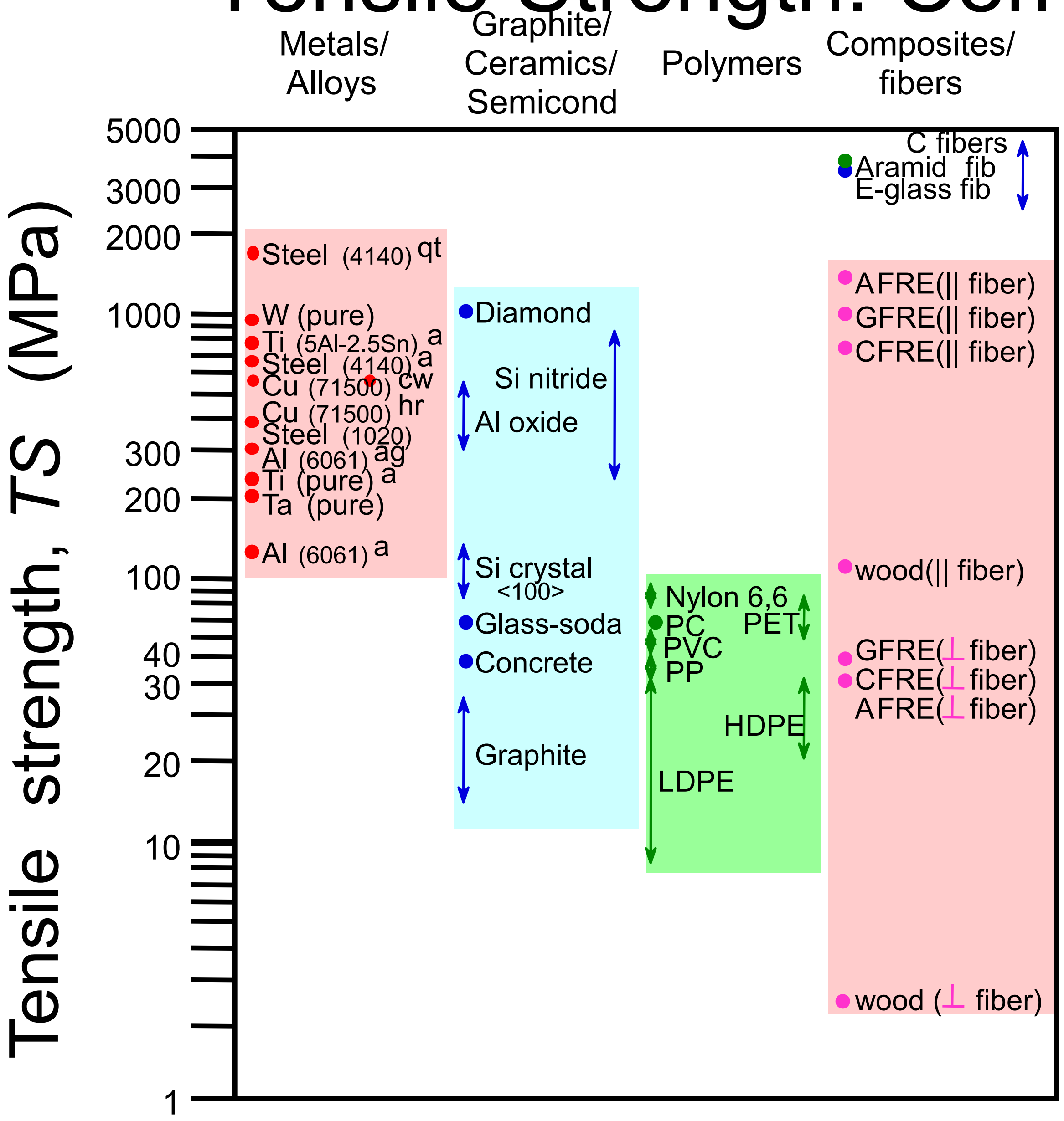
Necking

2. <u>Ultimate stress</u> (or <u>tensile strength</u>): the maximum stress which also indicates the overall strength of material

Ductile materials: Any materials that can be subjected to large strains before it ruptures – Necking

Brittle materials: Materials that exhibit little or no yielding before failure

Tensile Strength: Comparison

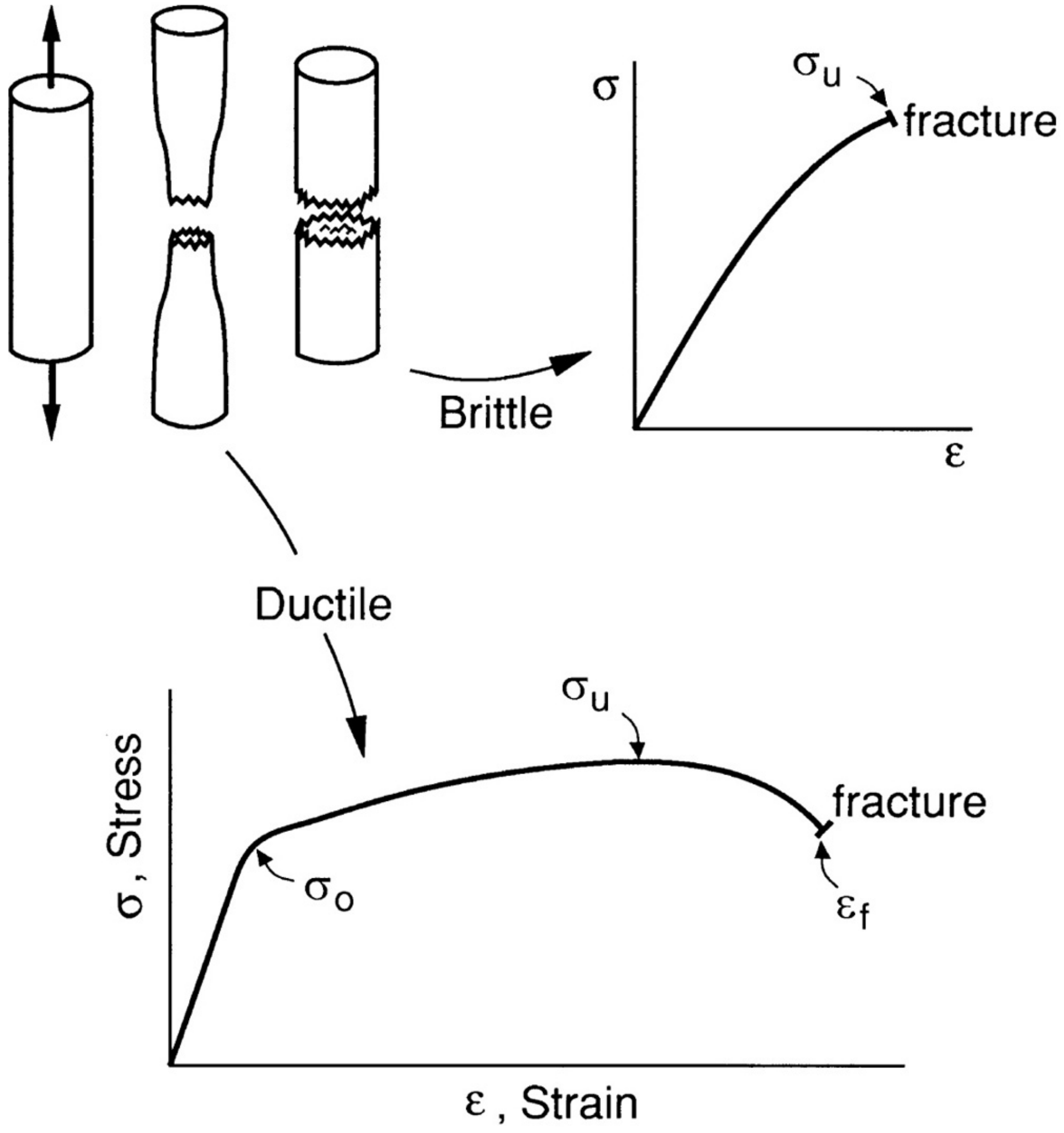


Room temperature values

Based on data in Table B4, *Callister & Rethwisch 3e.*

a = annealed
hr = hot rolled
ag = aged
cd = cold drawn
cw = cold worked
qt = quenched & tempered
AFRE, GFRE, & CFRE = aramid, glass, & carbon fiber-reinforced epoxy composites, with 60 vol% fibers.

Ductile and Brittle Materials have different Failures



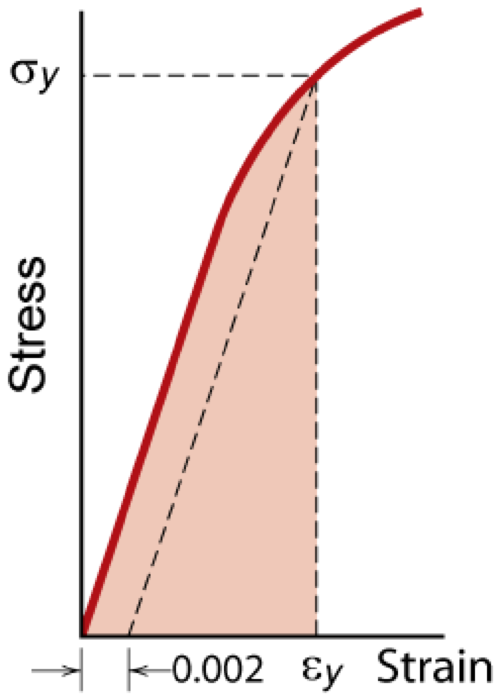
Mechanics of materials, R. C. Hibbeler

Material properties measured by tension tests

- Stiffness:
  - Elastic (Young's) modulus
- Strength:
  - Yield stress (σ_y,YS)
  - Ultimate tensile stress (σ_u ,UTS) or tensile strength (TS)
- Ductility:
  - Percent elongation
  - Percent reduction in area.

Resilience, U_r

- Ability of a material to store energy
  - Energy stored in **elastic region**



Adapted from Fig. 7.15,
Callister & Rethwisch 3e.

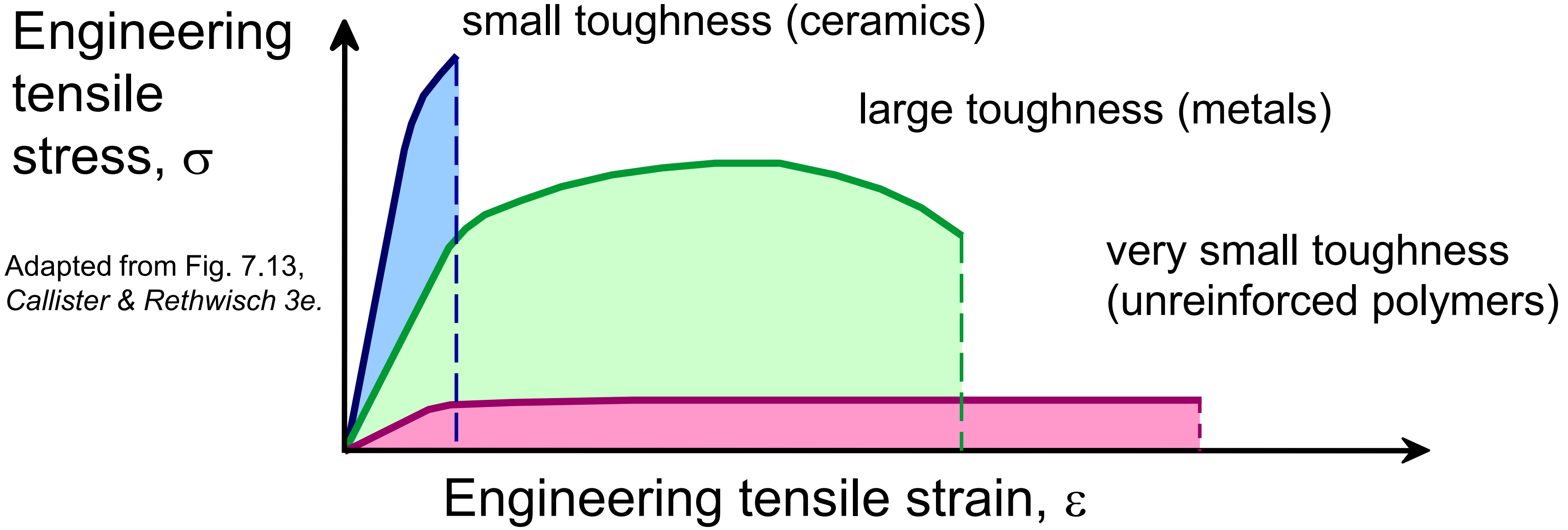
$$U_r = \int_0^{\varepsilon_y} \sigma d\varepsilon$$

If we assume a linear stress-strain curve this simplifies to:

$$U_r \cong \frac{1}{2}\sigma_y \varepsilon_y$$

Toughness, U_T

- Energy to break a unit volume of material
- Approximate by the area under the stress-strain curve.



Adapted from Fig. 7.13, *Callister & Rethwisch 3e.*

Brittle fracture: elastic energy
Ductile fracture: elastic + plastic energy

Toughness

- The supporting arms of a car bumper are an example of where toughness is of great value as a mechanical property.

- Key properties of a car bumper: elastic modulus, yield point, tensile strength and fracture toughness

Practice Calculations

- A cylindrical aluminum alloy test bar is subjected to a tension load of 900 *lbs*. The initial diameter of the sample is 0.357 *in* and the initial gage length 2 *in*. At fracture, the length change (elongation) is 0.335 *in*.

- Determine engineering <u>stress</u> and <u>strain</u> right before fracture.

Stress Strain Curve

Strain (mm/mm)	Stress (MPa)
0	0
0.00098	196
0.002008	401.6
0.002992	598.4
0.003996	799.2
0.005	900
0.01	940
0.022	990
0.056	1040
0.076	1070
0.096	1090
0.116	1100
0.136	1080
0.151	1040
0.166	950
0.181	800